



Refraction, Reflection & Polarisation

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Equation for the Intensity of Radiation



Your notes

Equation for the Intensity of Radiation

- Progressive waves transfer **energy**
- The amount of energy passing through a unit area per unit time is the **intensity** of the wave
- Therefore, the **intensity** is defined as **power per unit area**

Diagram illustrating the equation for intensity: $I = \frac{P}{A}$. The equation is shown with arrows pointing to the variables: **INTENSITY (Wm^{-2})** points to I , **POWER (W)** points to P , and **AREA (m^2)** points to A .

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Intensity is equal to the power per unit area

- The **area** the wave passes through is **perpendicular** to the direction of its **velocity**
- The **intensity** of a progressive wave is also **proportional** to its **amplitude squared** and **frequency squared**

Diagram illustrating the proportionality of intensity: $I \propto A^2$ and $I \propto f^2$. The equation is shown with arrows pointing to the variables: **INTENSITY (Wm^{-2})** points to I , ***PROPORTIONAL TO*** points to $\propto$, **AMPLITUDE (m)** points to A , and **FREQUENCY (Hz)** points to f .

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Intensity is proportional to the amplitude² and frequency²

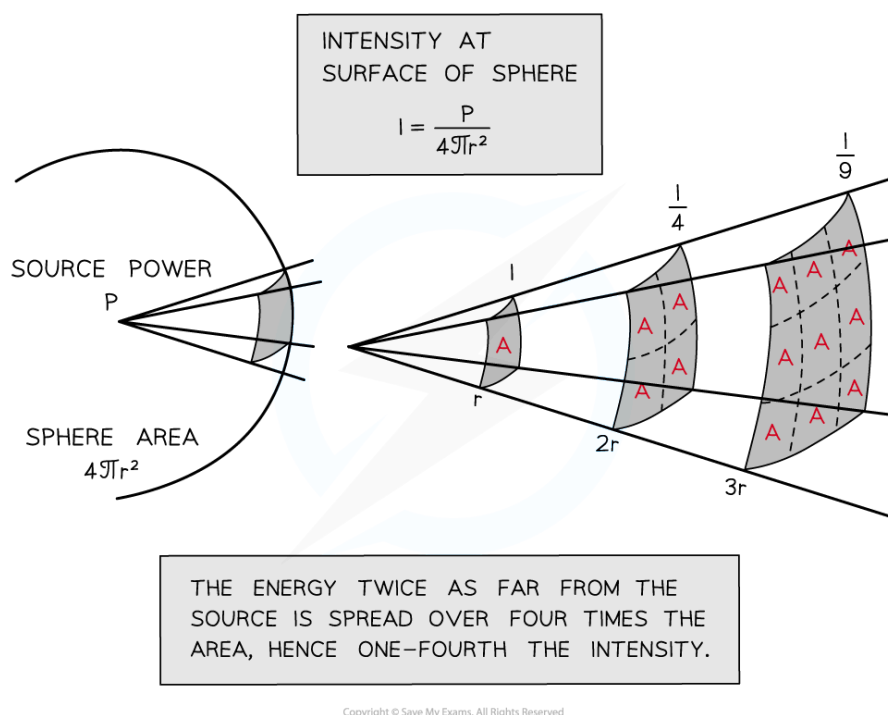
- This means, if the frequency or the amplitude is doubled, the intensity increases by a factor of 4 (2^2)

Spherical Waves

- A **spherical wave** is a wave from a point source that **spreads out equally** in all directions
- The area the wave passes through is the **surface area** of a sphere: $4\pi r^2$
- As the wave travels further from the source, the energy it carries passes through increasingly larger areas as shown in the diagram below:



Your notes



Intensity is proportional to the amplitude squared

- Assuming there's no absorption of the wave energy, the **intensity** / **decreases** with **increasing distance** from the source
- Note the intensity is proportional to $1/r^2$
 - This means when the source is twice as far away, the intensity is 4 times less
- The $1/r^2$ relationship is known in physics as the **inverse square law**

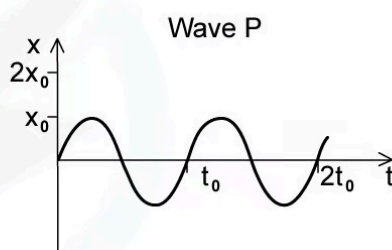
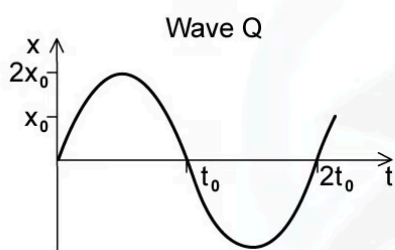


Worked Example

The intensity of a progressive wave is proportional to the square of the amplitude of the wave. It is also proportional to the square of the frequency. The variation with time of displacement x of particles when two progressive waves Q and P pass separately through a medium are shown on the graphs



Your notes



The intensity of wave Q is I_0 . What is the intensity of wave P?

Answer:

STEP 1

INTENSITY EQUATION

$$I \propto A^2$$

$$I \propto f^2$$

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STEP 2

CALCULATE HOW MUCH THE AMPLITUDE HAS INCREASED/DECREASED

WAVE P IS HALF THE AMPLITUDE OF WAVE Q

$$A_P = \frac{1}{2} A_Q$$

$$\text{WAVE P} = \frac{1}{4} I_0 \text{ OF WAVE Q}$$

STEP 3

CALCULATE HOW MUCH THE FREQUENCY HAS INCREASED/DECREASED

WAVE P IS DOUBLE THE FREQUENCY OF WAVE Q

$$f_P = 2 f_Q$$

$$\text{WAVE P} = 4 I_0 \text{ OF WAVE Q}$$

STEP 4

SUBSTITUTE BACK INTO INTENSITY EQUATION

$$I_0(P) \propto \left(\frac{1}{4} \times 4\right) I_0(Q)$$

$$\text{INTENSITY OF WAVE P} = \text{INTENSITY OF WAVE Q} = I_0$$

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Examiner Tips and Tricks

The key concept with intensity is that it has an inverse square relationship with distance (not a linear one). This means the energy of a wave decreases very rapidly with increasing distance

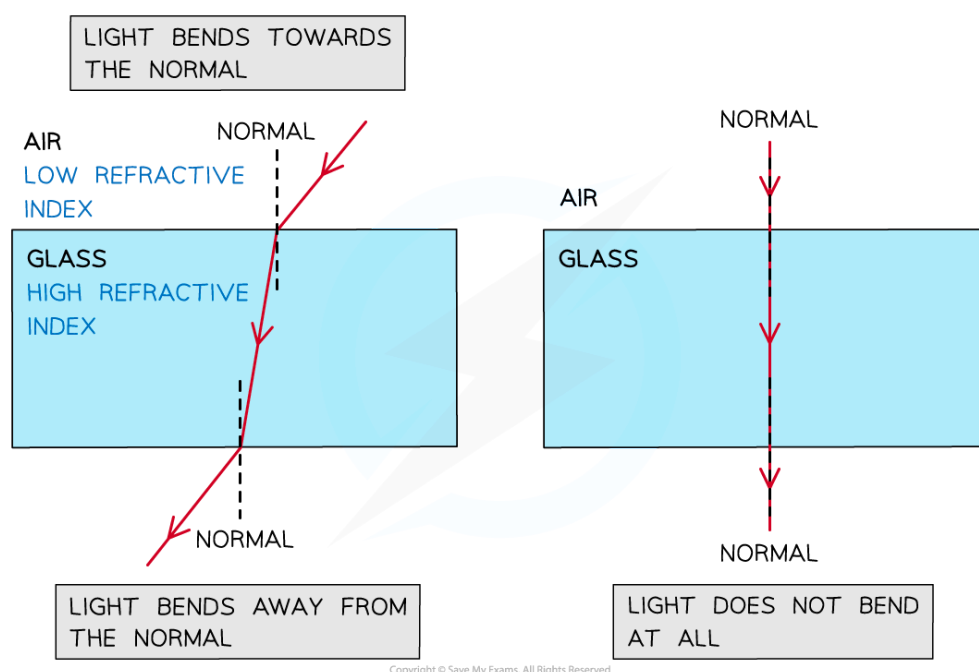


Your notes



Refraction & Refractive Index

- Refraction occurs when light passes a boundary between two different transparent media
- At the boundary, the rays of light undergo a **change in direction** and a **change in speed**
- The change in direction is caused by the change in speed
 - Entering a **more dense** medium **slows** the light down and it bends **towards** the normal
 - In the denser medium there are more particles closer together providing more friction to the passing of the light through the material
 - Entering a **less dense** medium **speeds** the light up and it bends **away** from the normal
 - When passing **along the normal** (perpendicular) the light does **not** direction
 - Its speed does still change, as it is passing through a medium with a different refractive index



Refraction of light through a glass block

Calculating Refractive Index

- The refractive index, n , is a property of a material which measures how much light slows down when passing through it

$$n = \frac{c}{v}$$



Your notes

- Where:
 - c = the speed of light in a vacuum (m s^{-1})
 - v = the speed of light in a substance (m s^{-1})
- Light travels at different speeds within different substances depending on their refractive index
 - A material with a high refractive index is called **optically dense**, such material causes light to travel **slower**
- Since the speed of light in a substance will always be less than the speed of light in a vacuum, the value of the n is **always greater than 1**
- In calculations, the refractive index of air can be taken to be approximately 1
 - This is because light does not slow down significantly when travelling through air (as opposed to travelling through a vacuum)

Snell's Law

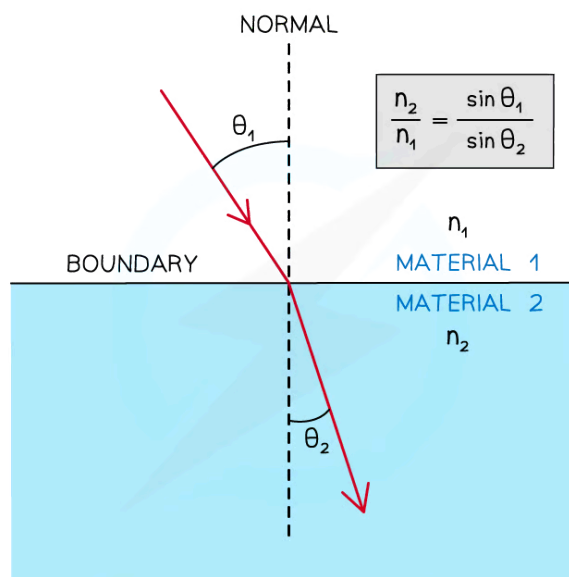
- Snell's law relates the **angle of incidence** to the **angle of refraction**, it is given by:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

- Where:
 - n_1 = the refractive index of material 1
 - n_2 = the refractive index of material 2
 - θ_1 = the angle of incidence of the ray in material 1 ($^\circ$)
 - θ_2 = the angle of refraction of the ray in material 2 ($^\circ$)



Your notes



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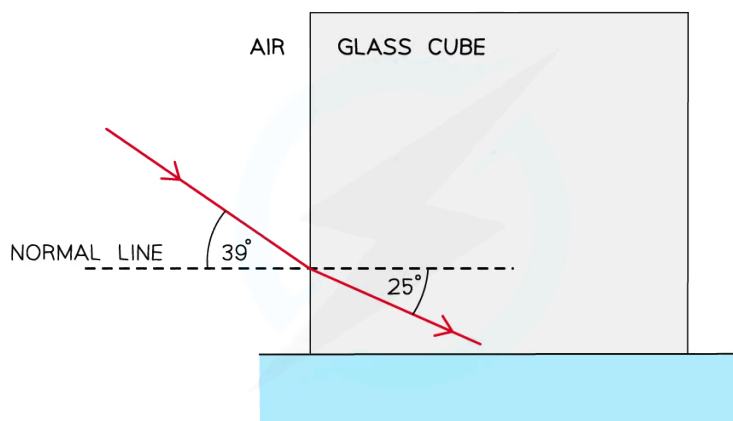
Snell's Law is used to find the refractive indices or the angles to the normal at a boundary

- θ_1 and θ_2 are always taken from the **normal**
- Material 1 is always the material in which the ray goes through **first**
- Material 2 is always the material in which the ray goes through **second**



Worked Example

A light ray is directed at a vertical face of a glass cube. The angle of incidence at the vertical face is 39° and the angle of refraction is 25° as shown in the diagram.



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Show that the refractive index of the glass is about 1.5.

Answer:



Your notes

Step 1: Write down the known quantities

- Refractive index of air, $n_1 = 1$
- Refractive index of glass = n_2
- Angle of incidence, $\theta_1 = 39^\circ$
- Angle of refraction, $\theta_2 = 25^\circ$

Step 2: Write out Snell's Law

$$\frac{n_2}{n_1} = \frac{\sin \theta_1}{\sin \theta_2}$$

Step 3: Calculate the refractive index of glass

$$n_2 = \frac{\sin 39}{\sin 25} = 1.489 = 1.5 \text{ (2 s.f.)}$$

- Note that in a "show that" question, the answer should be to at least one more significant figure than the value given in the question



Examiner Tips and Tricks

Always double-check if your calculations for the refractive index are greater than 1. Otherwise, something has definitely gone wrong in your calculation!

The refractive index of air will not be given in the question. Always assume that $n_{\text{air}} = 1$.

Always check that the angle of incidence and refraction are the angles between the normal and the light ray. Remember the normal line is not really there – it has been drawn in to give you a place to measure from.



Critical Angle

- As the angle of incidence at the boundary between a more dense and a less dense medium is increased, the angle of refraction also increases until it gets to 90°
- When the angle of refraction is exactly 90° the light is refracted along the boundary (if the boundary is straight)
 - At this point, the angle of incidence is known as the **critical angle C**
 - This can only occur when light passes from a **more dense** to a **less dense** material
- This angle can be found using the formula:

$$\sin(C) = \frac{1}{n}$$

- This can easily be derived from Snell's law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

- Where:
 - $\theta_1 = C$
 - $\theta_2 = 90^\circ$
 - $n_1 = n$
 - $n_2 = 1$ (air)

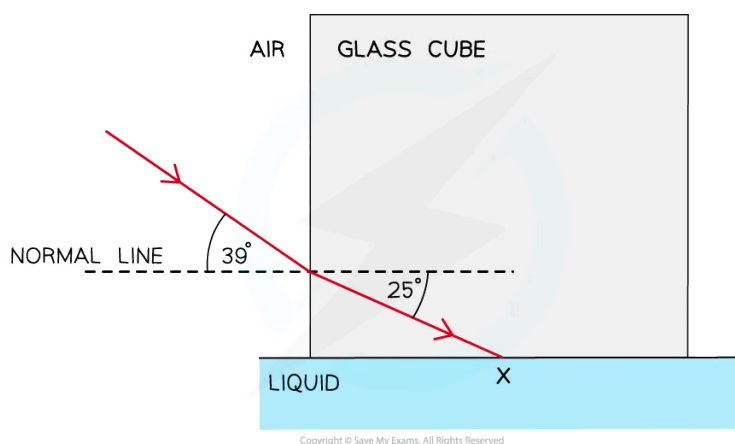


Worked Example

A glass cube is held in contact with a liquid and a light ray is directed at the vertical face of the cube. The angle of incidence at the vertical face is 39° and the angle of refraction is 25° as shown in the diagram. The light ray is totally internally reflected at **X**. The refractive index of the glass cube is 1.45 and the refractive index of the liquid is 1.32.



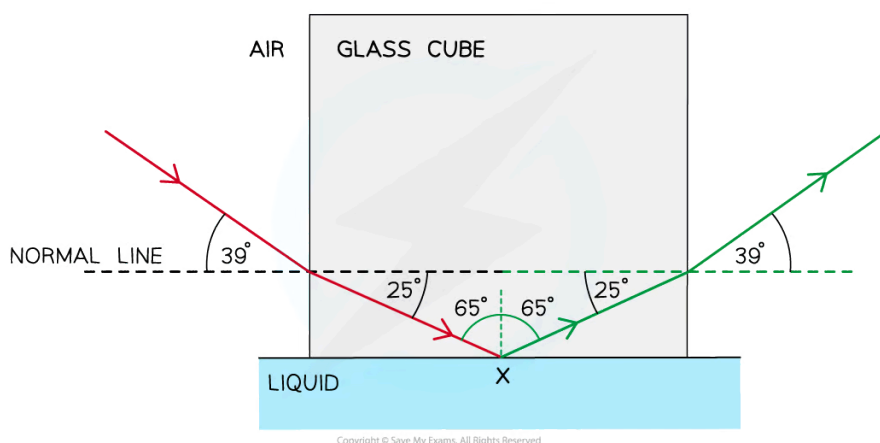
Your notes



- a) Complete the diagram to show the path of the ray beyond X
- b) Calculate the critical angle for the ray at the glass-liquid boundary

Answer:

- a) Complete the diagram to show the path of the ray beyond X



Step 1: Draw the reflected angle at the glass-liquid boundary

- When a light ray is reflected, the angle of incidence = angle of reflection
- Therefore, the angle of incidence (and reflection) is $90^\circ - 25^\circ = 65^\circ$

Step 2: Draw the refracted angle at the glass-air boundary

- At the glass-air boundary, the light ray refracts **away** from the normal
 - Due to the reflection, the light rays are symmetrical to the other side
- b) Calculate the critical angle for the ray at the glass-liquid boundary:

Step 1: Recall Snell's Law and rearrange to make critical angle the subject

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$



Your notes

$$\sin(C) = \frac{n_2}{n_1} \sin(\theta_2)$$

Step 2: Substitute in the known quantities

- n_1 = refractive index of glass cube = 1.45
- n_2 = refractive index of liquid = 1.32
- $\theta_1 = C$
- $\theta_2 = 90^\circ$ (The angle of refraction is 90° when at the critical angle)

$$\sin(C) = \frac{n_2}{n_1} \sin(\theta_2)$$

$$C = \sin^{-1}\left(\frac{1.32}{1.45}\right)$$

Step 3: Calculate the critical angle

$$C = 65.55^\circ$$



Examiner Tips and Tricks

Always draw ray diagrams with a ruler, and make sure you're comfortable calculating unknown angles. The main rules to remember are:

- Angles in a right angle add up to 90°
- Angles on a straight line add up to 180°
- Angles in any triangle add up to 180°

For angles in parallel lines, such as alternate and opposite angles, take a look at the OCR GCSE maths revision notes '7.1.1 Angles in Parallel Lines'



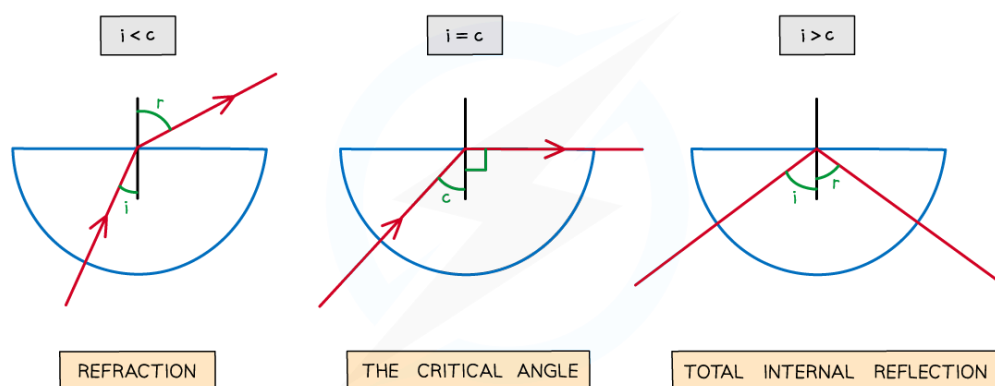
Total Internal Reflection

- Total internal reflection (TIR) occurs when:

The angle of incidence is greater than the critical angle and the incident refractive index n_1 is greater than the refractive index of the material at the boundary n_2

- Therefore, the **two** conditions for total internal reflection are:

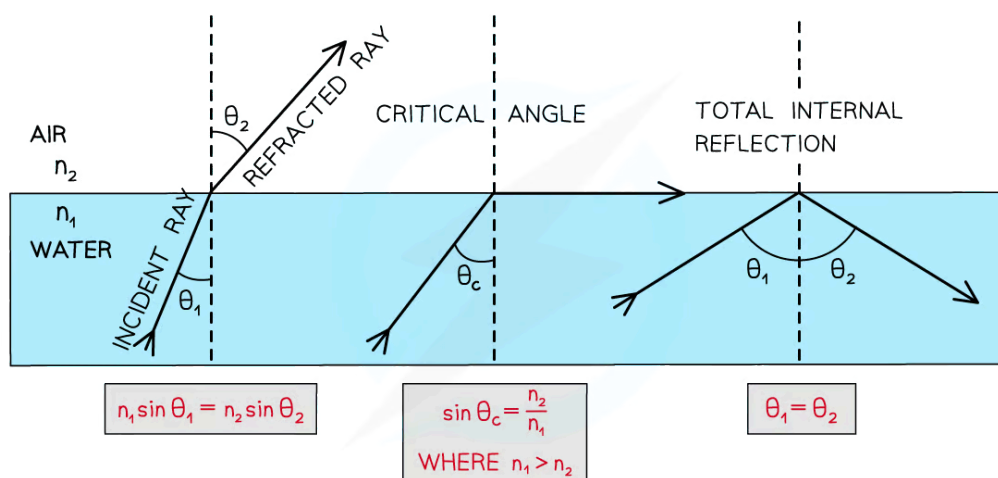
- The angle of incidence, $\theta_1 >$ the critical angle, C
- Refractive index $n_1 >$ refractive index n_2 (air)



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Diagram showing refraction, the critical angle and total internal reflection

- Two conditions are necessary for total internal reflection to occur:
 - The light must be going from a **more dense medium** into a **less dense one**
 - The **angle of incidence** must be **greater** than the **critical angle**



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Examiner Tips and Tricks

If asked to name the phenomena make sure you **give the whole name** - Total Internal Reflection. Remember: Total Internal Reflection occurs when going from a **more dense to a less dense** material and **ALL** of the light is reflected. If asked to explain what is meant by the **critical angle**, you can draw the diagram above (showing the three semi-circular blocks).



Your notes



Measuring Refractive Index

Aim of the Experiment

- To investigate the **refraction** of light through a **perspex block**

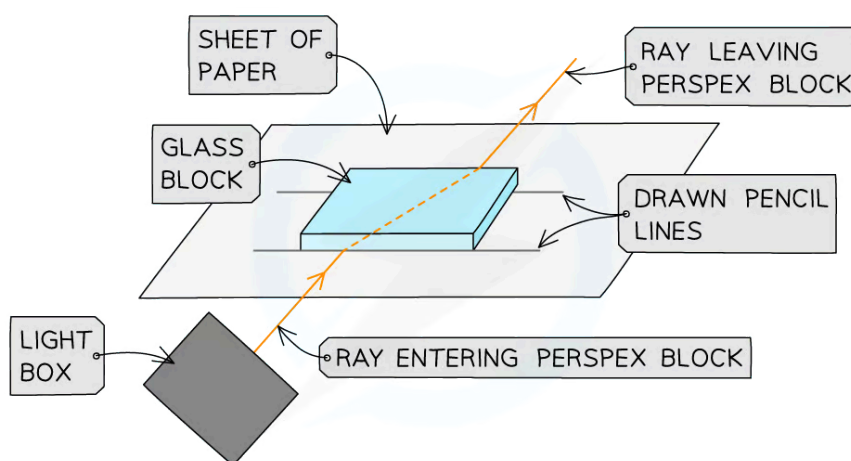
Equipment

- Ray Box - to provide a narrow beam of light to refract through the perspex box
- Protractor - to measure the light beam angles
- Sheet of paper - to mark with lines for angle measurement
- Pencil - to make perpendicular line and angle lines on paper
- Ruler - to draw straight lines on the paper
- Perspex block - to refract the light beam

Variables

- Dependent variable = angle of refraction, r
- Control variables:
 - Use of the same perspex block
 - Width of the light beam
 - Same frequency / wavelength of the light

Method



Apparatus to investigate refraction

1. Place the **perspex block** on a **sheet of paper**, and draw around it using a pencil



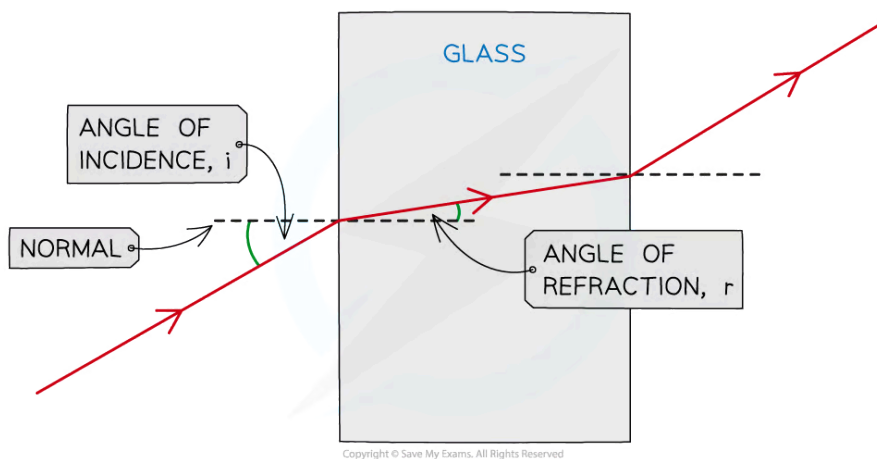
2. Switch on the **ray box** and direct a beam of light at the **side face** of the block
3. **Mark** on the paper with a small 'x':
 - A point on the ray close to the ray box
 - The point where the ray enters the block
 - The point where the ray exits the block
 - A point on the exit light ray which is a distance of about 5 cm away from the block
4. **Draw** a dashed line **normal** (at right angles) to the outline of the block
5. Remove the block and **join the points** marked 'x' with three straight lines
6. Replace the block within its outline and **repeat** the above process for a ray striking the block at **different angles** of incidence
 - An example of the data collection table is shown below:

Angle of incidence, i	Angle of refraction, r
30°	
45°	
60°	
90°	

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Analysis of Results

- i and r are always measured from the **normal**
- For light rays **entering** perspex block, the light ray refracts **towards** the central line:
$$i > r$$
- For light rays **exiting** the perspex block, the light ray refracts **away** from the central line:
$$i < r$$
- When the angle of incidence is 90° to the perspex block, the light ray does **not** refract, it passes straight through the block:
$$i = r$$
- If the experiment was carried out correctly, the angles should follow the pattern, as shown below:



How to measure the angle of incidence and angle of refraction

Safety Considerations

- The ray box light gets **hot** and could **burn** if touched
 - Run burns under cold running water for at least five minutes
- Looking directly into the light may **damage** the **eyes**
 - Avoid looking directly at the light
 - Stand behind the ray box during the experiment
- Keep all **liquids away** from the electrical equipment and paper
- Take care using the perspex
 - Damage to the perspex block can affect the outcome of the experiment



Examiner Tips and Tricks

In your examination you could be asked about the method for this experiment or given a set of results and asked how accurate they are or how they can be improved.

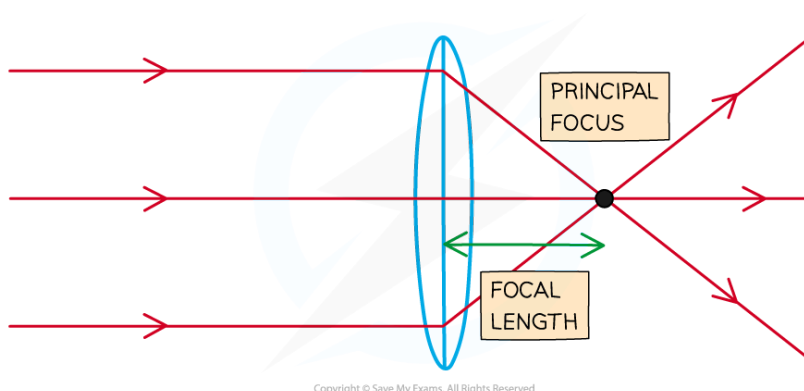


Converging & Diverging Lenses

- A lens is a piece of equipment that forms an image by **refracting** light
- There are two types of lens:
 - Convex
 - Concave

Convex Lenses

- In a convex lens, parallel rays of light are brought to a focus by refraction
 - This point is called the **principal focus**
- This lens is sometimes referred to as a **converging** lens
- The distance from the lens to the principal focus is called the **focal length**
 - This depends on how **curved** the lens is
 - The **more** curved the lens, the **shorter** the focal length



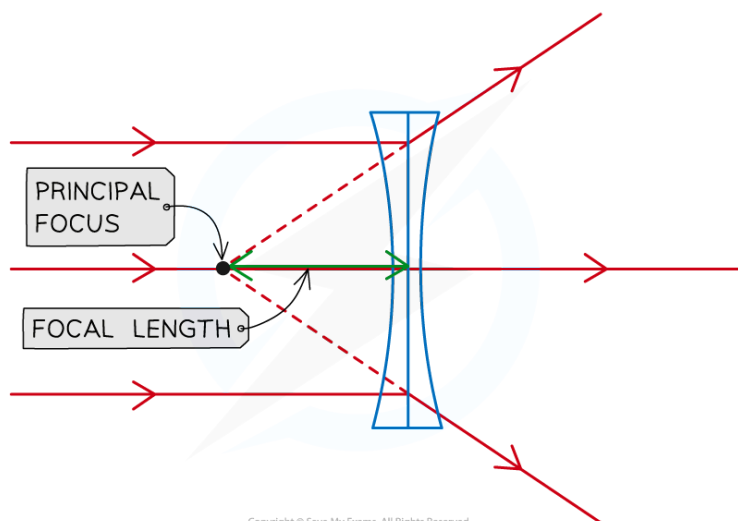
The focal length is the distance from the lens to the principal focus

Concave Lenses

- In a concave lens, parallel rays of light are made to diverge (spread out) from a point
 - This lens is sometimes referred to as a **diverging** lens
- The principal focus is now the point from which the rays appear to diverge from



Your notes



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Parallel rays from a concave lens appear to come from the principal focus



Examiner Tips and Tricks

To remember which lens is converging or diverging, think of the following: Convex lens = Converging

To remember which lens is which, a concave lens goes in at the middle, like a cave. Okay, not a very exciting cave, but all the same...



Using Ray Diagrams

Describing images formed by lenses

- Images are described using three concepts
- They are either:
 - **Real** or **virtual**
 - **Bigger** than, the **same size** as or **smaller** than the object
 - **Inverted** or the **same way up** as the object
- A **real image** is one formed by the **convergence of rays** of light (the rays meet)
 - A real image can be projected onto a screen
- A **virtual image** is seen but not formed on a screen
 - The rays of light have not met, they have been perceived by the eye
 - An image viewed through a **magnifying glass** is a virtual image
- Lenses can be used to form images of objects placed in front of them
- The location (and nature) of the image can be found by drawing a ray diagram:

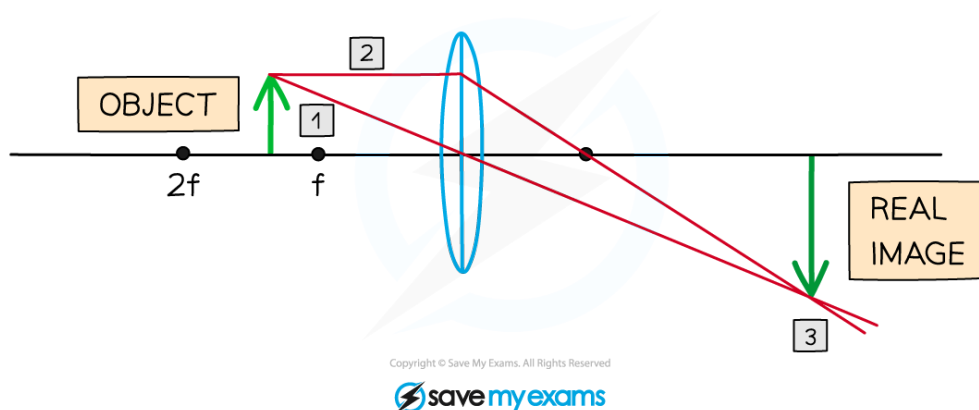


Diagram showing the formation of a real image by a lens

Drawing a Ray Diagram

1. Start by drawing a **ray going from the top** of the object **through the centre** of the lens.
This ray will continue to travel in a straight line
2. Next draw a **ray going from the top of the object**, travelling **parallel to the axis** to the lens. When this ray emerges from the lens it will travel directly towards the principal focus



3. The image is found at the point where the two rays meet

- The above diagram shows the image that is formed when the object is placed at a distance between one focal length (f) and two focal lengths ($2f$) from the lens
- In this case, the image is:
 - **Real**
 - **Enlarged**
 - **Inverted**
- The following diagram shows what happens when the object is more distanced – further than twice the focal length ($2f$) from the lens:

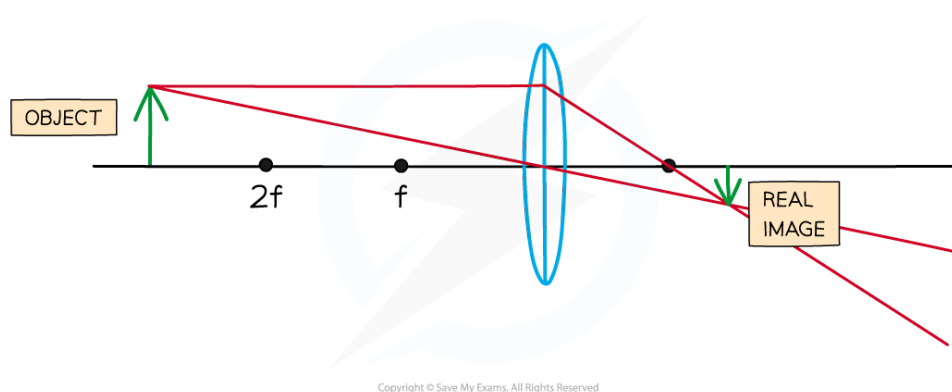


Diagram showing the formation of a real image by a lens with the object at distance

- In this case the image is:
 - **Real**
 - **Diminished (smaller)**
 - **Inverted**
- If the object is placed at exactly twice the focal length ($2f$) from the lens:

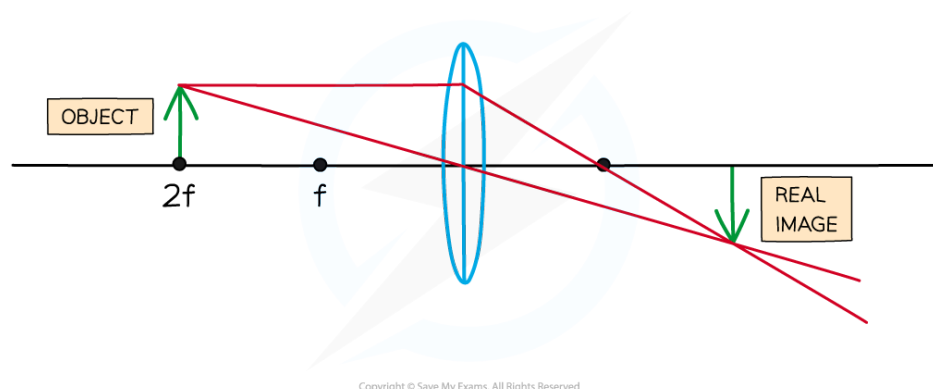


Diagram showing the formation of a real image with the object at $2f$

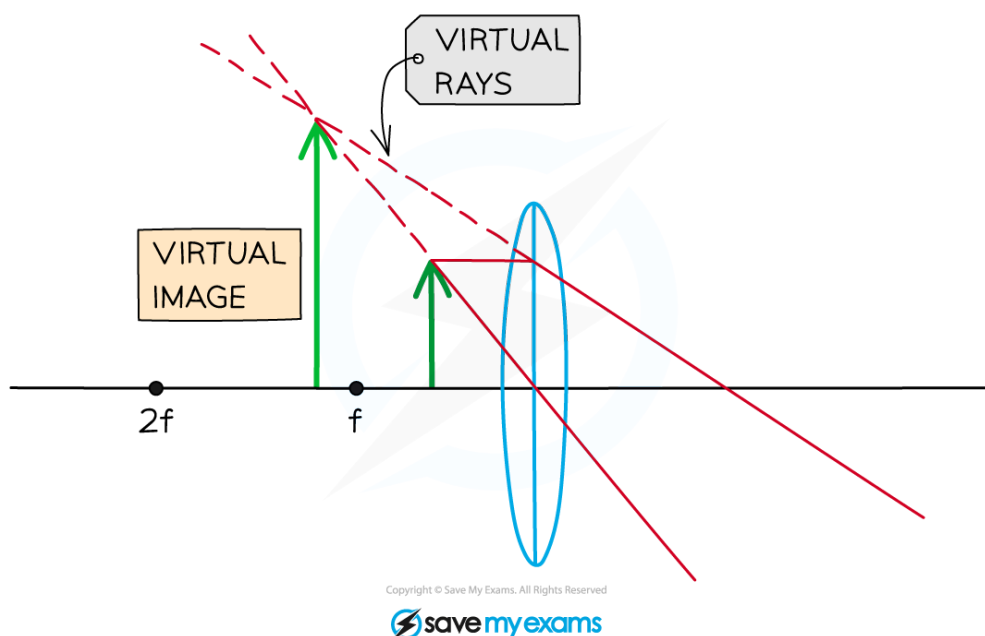


Your notes

- In this case the image is:
 - Real
 - Same size as the object
 - Inverted

Magnifying glasses

- If the object is placed **closer** to the **lens** than the **focal length**, the emerging **rays diverge** and a **virtual image** is formed
- When viewed from the right-hand side of the lens, the emerging rays appear to come from a point on the left. This point can be found by extending the rays backwards (creating virtual rays)
- A **virtual image** will be seen at the point where these virtual rays cross



A virtual image is formed by the divergence of rays from a point

- In this case the image is:
 - Virtual
 - Enlarged
 - Upright
- Using a lens in this way allows it to be used as a **magnifying glass**
- When using a magnifying glass, the lens should always be held close to the object



Examiner Tips and Tricks

It is important to understand how the images are formed in both examples, as well as the type of image formed. You should practice drawing accurate lens diagrams. They are harder than they look!

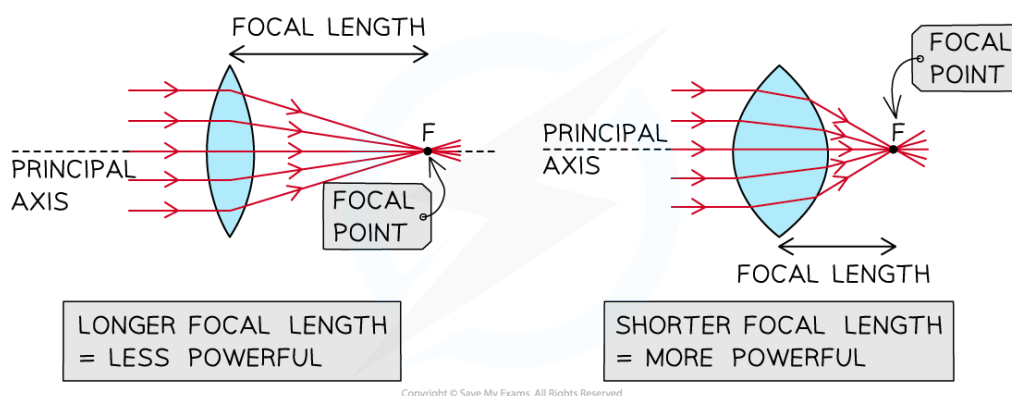


Your notes



Power of a Lens

- The **power** of a lens is a measure of its ability to **refract** light
 - The **more** refraction a lens causes, the **higher** its power
- The **power** of a lens measures how strongly it focuses the light
 - The **more** curved the lens, the **shorter** the focal length
 - The **shorter** the focal length, the **greater** the power of the lens
- The **power** of a lens is related to:
 - The **focal length** of the lens
 - The **shape** of the lens



The power of a lens depends on its focal length

Calculate the power of a lens

- The power of a lens, P is calculated using the following equation:

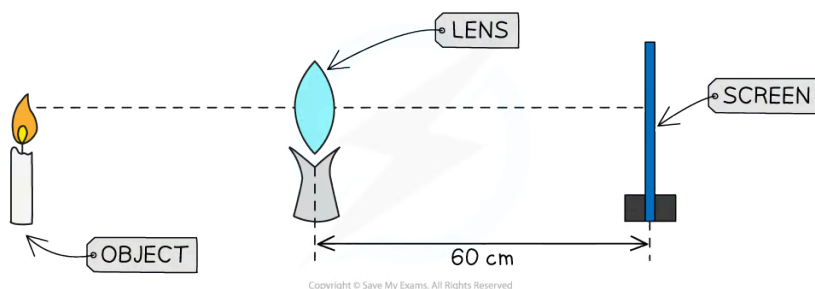
$$P = \frac{1}{f}$$

- Where:
 - P = power (dioptries, D)
 - f = focal length of the lens (m)
- **Power** is **inversely proportional** to **focal length**
- For a **concave mirror**, where the focal length is negative, power has a negative value



Worked Example

A lens is set up between an object candle and a screen. All the equipment is arranged perpendicular to the desk. The image of the candle is in focus when the screen is positioned 60 cm from the lens.



Determine the power of the lens.

Answer:

Step 1: Write the known quantities in S.I. units

- Image is in focus, so the screen is at the focal point
- Focal length, $f = 60 \text{ cm} = 0.6 \text{ m}$

Step 2: Write the equation for power and substitute the values

$$P = \frac{1}{f} = \frac{1}{0.6} = 1.67$$

Step 3: Give the full answer, to correct significant figures and with units

The power of the lens, $P = 1.7 \text{ D}$



Examiner Tips and Tricks

The explanations above relate to life in a common sense way. Stronger, more powerful reading glasses are used by people who have the weakest eye sight. They need lenses to do more of the focusing for them.



Your notes

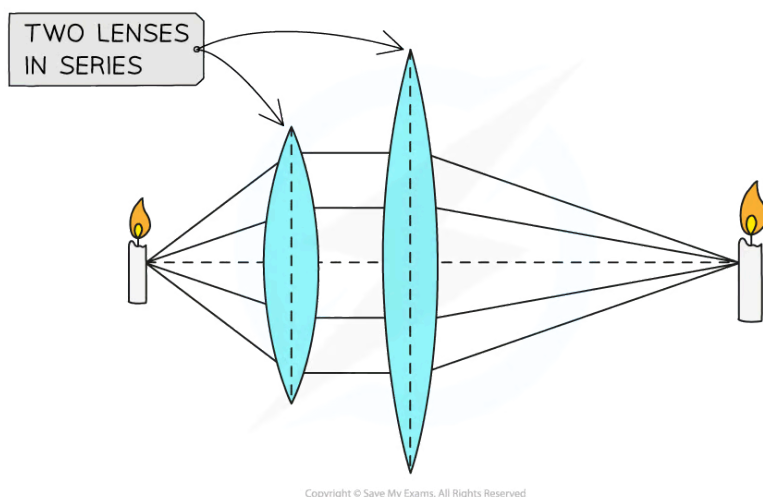


Thin Lenses in Combination

- When **multiple lenses** are used in **series** (one after the other) this is called a **compound lens**
- To find the **total power** of a compound lens, find the **sum of the powers** of the **individual lenses**

$$P_{\text{Total}} = P_1 + P_2 + \dots P_n$$

- Where:
 - P_{Total} = Total power of all the lenses (diopetre, D)
 - P_1 = power of lens 1 (diopetre, D)
 - P_2 = power of lens 2 (diopetre, D)



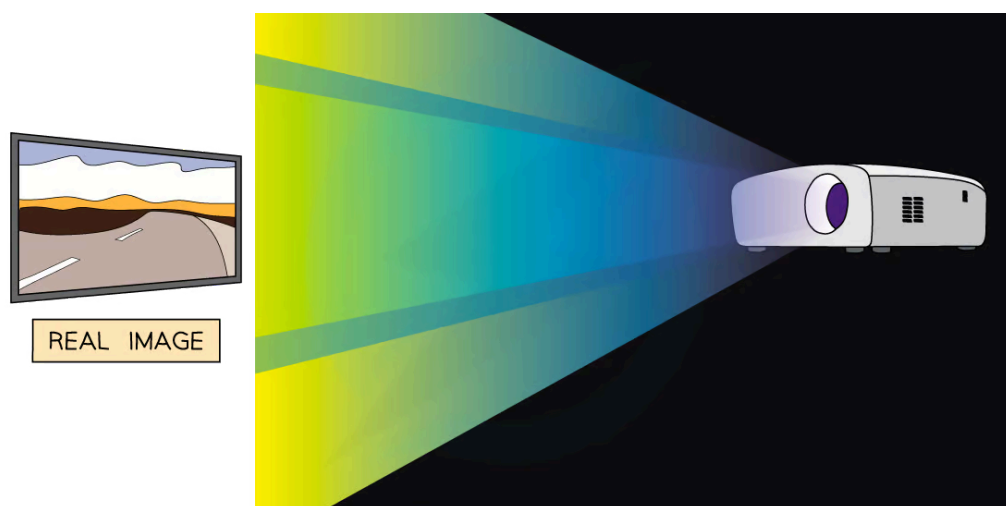
- In this case the image is:
 - Virtual
 - Diminished
 - Upright
- The lenses should be:
 - Arranged so that their principle axes line up
 - Touching or very close



Real & Virtual Images

Real Images

- Are formed when light rays from a point on an object **pass through another point** in space
 - The light rays are **really there**
- Can be **formed on a screen**
- Are seen in ray diagrams at the point where **rays cross**
- Examples include:
 - Pictures projected onto a wall or screen
 - The image formed on the retina



A real image can be projected onto a screen

Virtual Images

- Virtual images are formed when light rays from a point on an object **appear** to have come from another point in space
- The light rays are **not really** where the image appears to be
- The image **cannot be formed on a screen**
- Examples include:
 - Images seen through a **magnifying glass**
 - All** images formed by a **diverging** (concave) lens

- Reflections in a mirror



Your notes

A reflection in a mirror is an example of a virtual image

Ray diagrams

- Ray diagrams can be used to show whether an image will be real or virtual
- If the rays from the object naturally cross the image will be real
- If the rays have to be extended backwards to make them cross, the image is virtual
 - To signal virtual rays they are drawn as dashed lines rather than solid ones

Ray diagram for a real image in a converging lens

1. Start by drawing a ray going from the top of the object through the centre of the lens.
This ray will continue to travel in a straight line
2. Next draw a ray going from the top of the object, travelling parallel to the axis to the lens.
When this ray emerges from the lens it will travel directly towards the principal focus
3. The image is found at the point where the above two rays meet

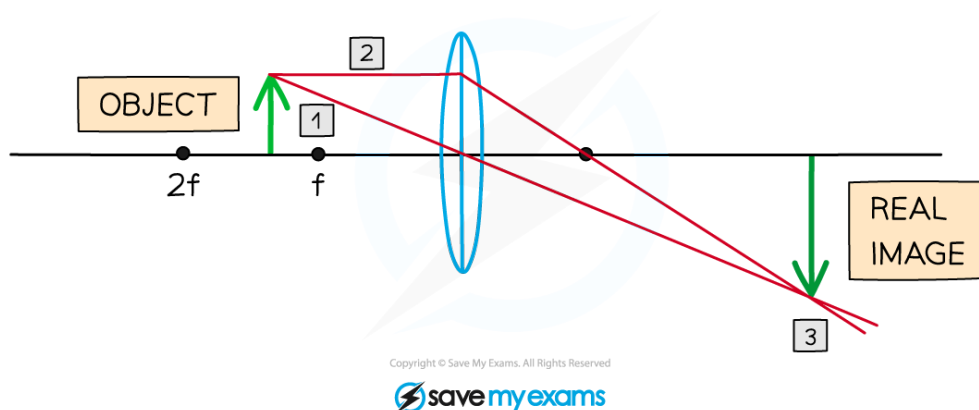


Diagram showing the formation of a real image by a lens

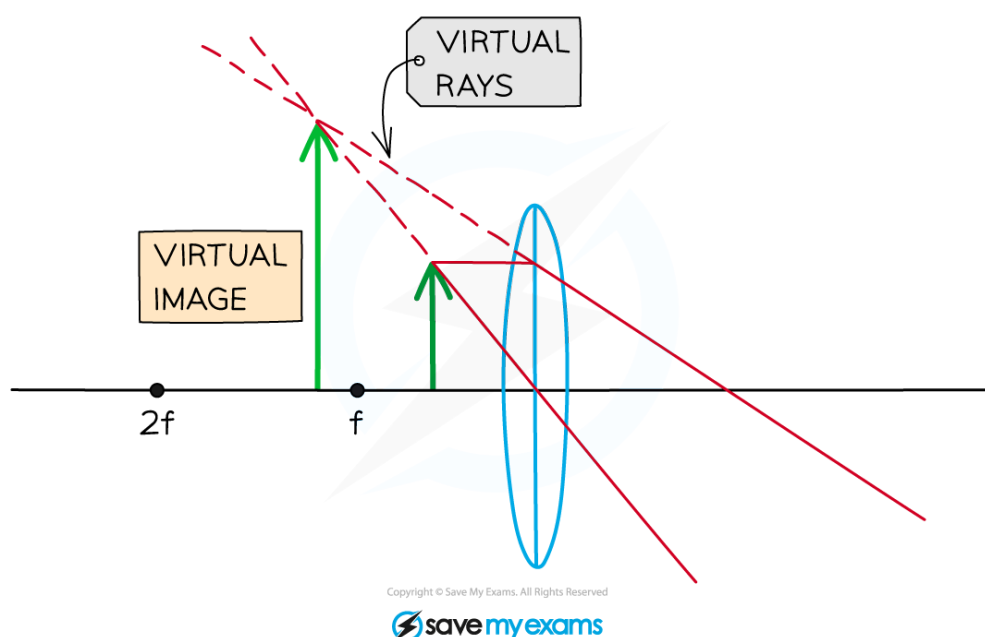


Your notes

- The above diagram shows the image that is formed when the object is placed at a distance between one focal length (f) and two focal lengths ($2f$) from the lens
- In this case, the image is:
 - Real
 - Enlarged
 - Inverted

Ray diagram for a virtual image in a converging lens

- If the object is placed closer to the lens than the focal length, the emerging rays diverge and a real image is no longer formed
- When viewed from the right-hand side of the lens, the emerging rays appear to come from a point on the left. This point can be found by extending the rays backwards (creating virtual rays)
- A virtual image will be seen at the point where these virtual rays cross



A virtual image is formed by the divergence of rays from a point

- In this case the image is:
 - Virtual
 - Enlarged
 - Upright
- Using a lens in this way allows it to be used as a **magnifying glass**

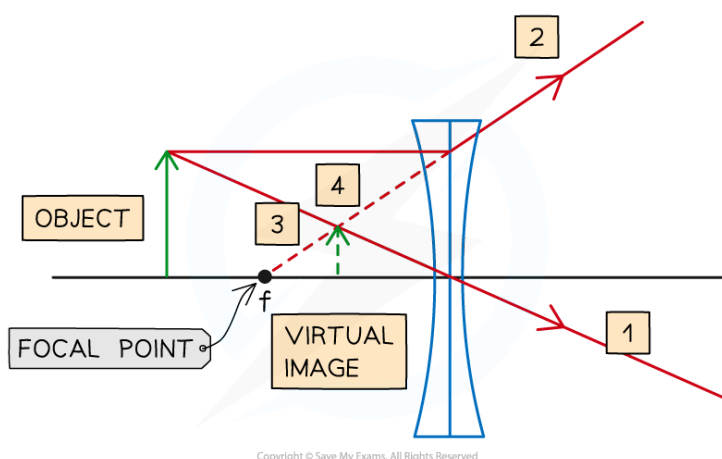


Your notes

- When using a magnifying glass, the lens should always be held close to the object

Ray diagram for a virtual image in a diverging lens

- The image formed by a diverging lens is always virtual
- To draw this diagram draw two rays from the top of the object
 - One ray passes through the centre of the lens with no refraction
 - The second is drawn parallel to the principal axis until it meets the centre of the lens
 - The ray refracts through the principal focus
- To make the rays cross the line will need to be extended, forming a virtual meeting point



Concave lenses only produce virtual images

- In this case the image is:
 - Virtual
 - Diminished
 - Upright



The Lens Equation

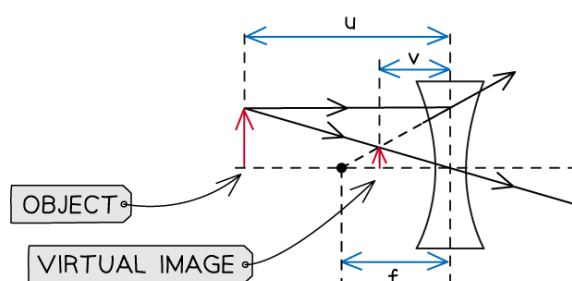
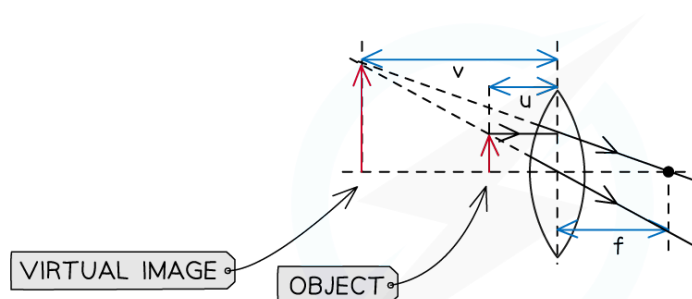
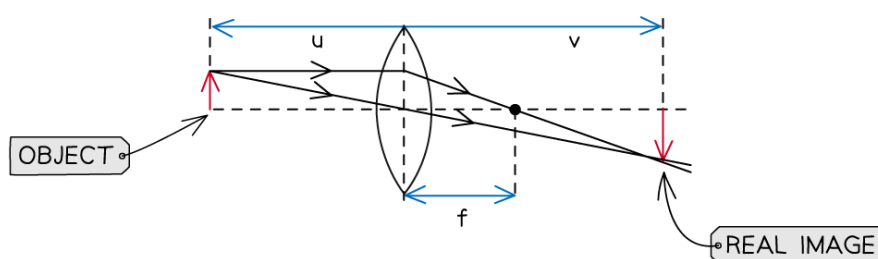
The Lens Equation

- This equation can be applied to all **thin converging** and **diverging** lenses
- The equation relates the **focal length** of the lens to the **distances** from the **lens** to the **image** and the **object**

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

- Where:

- f = focal length (m)
- v = image distance from lens (m)
- u = object distance from lens (m)



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- This equation only works for **thin** converging or diverging lenses
 - The values are **positive** if the image is **real**



Your notes



Worked Example

A student investigates the focal length of a thin lens by using it to project an image onto a screen.

The object is set 50.0 cm from the centre of the lens and the screen moved back and forth until the inverted image is sharp. This position is found to be 75.0 cm from the lens.

Determine the focal length of the lens.

Answer:

Step 1: Write the known values

- Distance from object to lens, $u = 50.0$ cm
- Distance from image to lens, $v = 75.0$ cm

Step 2: Write the equation and substitute in the values

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} = \frac{1}{50} + \frac{1}{75} = \frac{1}{30}$$

$$f = 30 \text{ cm}$$



Examiner Tips and Tricks

It is easy to forget the last step in a calculation like this one. Remember that you are calculating not f , and that you need to take the **reciprocal** of your answer.



Magnification

Magnification as a Ratio of Heights

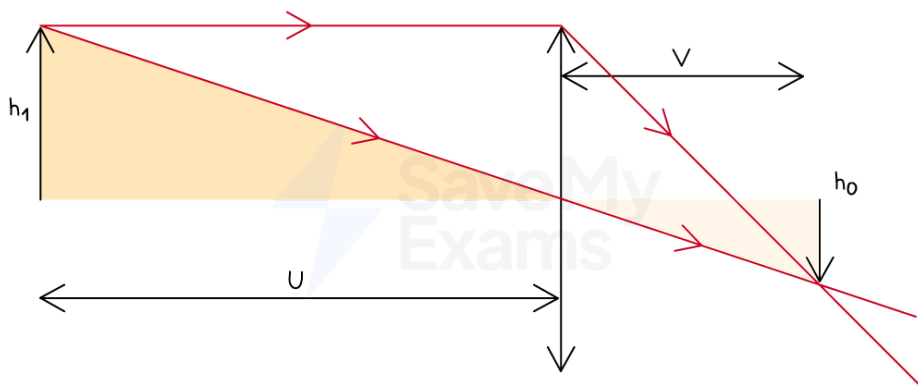
- Magnification means how much larger the image is than the object
 - This is the ratio of the image/object height

$$m = \frac{h_i}{h_o}$$

- Where:
 - m = magnification
 - h_i = image height (m)
 - h_o = object height (m)

Magnification as a Ratio of Distances

- A diagram of an object and its real image will produce similar triangles
 - Therefore, the ratio of magnification is also represented by comparing distance from the lens to the object and the image

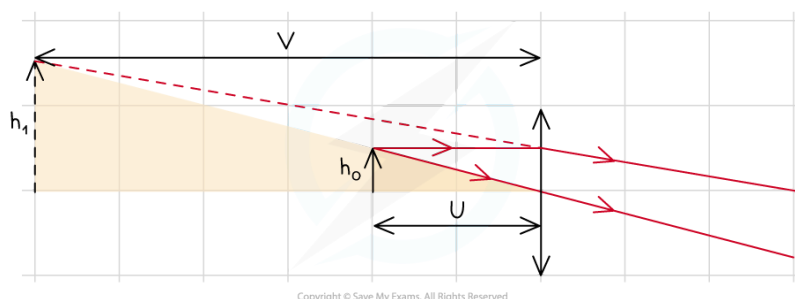


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- This also works for virtual images



Your notes



$$m = \frac{v}{u}$$

- Where:
 - m = magnification
 - v = distance from lens to image (m)
 - u = distance from lens to object (m)
- Since magnification is a **ratio**, it has **no units**



Worked Example

A magnifying glass has a focal length of 15 cm. It is held 5 cm away from a component which is being examined.

Determine the magnification of the image.

Answer:

Step 1: Write the known values

- Focal length, $f = 15$ cm
- Distance between object and lens, $u = 5$ cm

Step 2: Use the lens formula and rearrange to make v the subject

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$v = \left(\frac{1}{f} - \frac{1}{u} \right)^{-1} = \left(\frac{1}{15} - \frac{1}{5} \right)^{-1} = -7.5 \text{ cm}$$

- The negative sign indicates a virtual image (expected for a magnifying glass) and is ignored for the next step

Step 3: Use the magnification formula to find the magnification of the image

$$m = \frac{v}{u} = \frac{7.5}{5} = 1.5$$

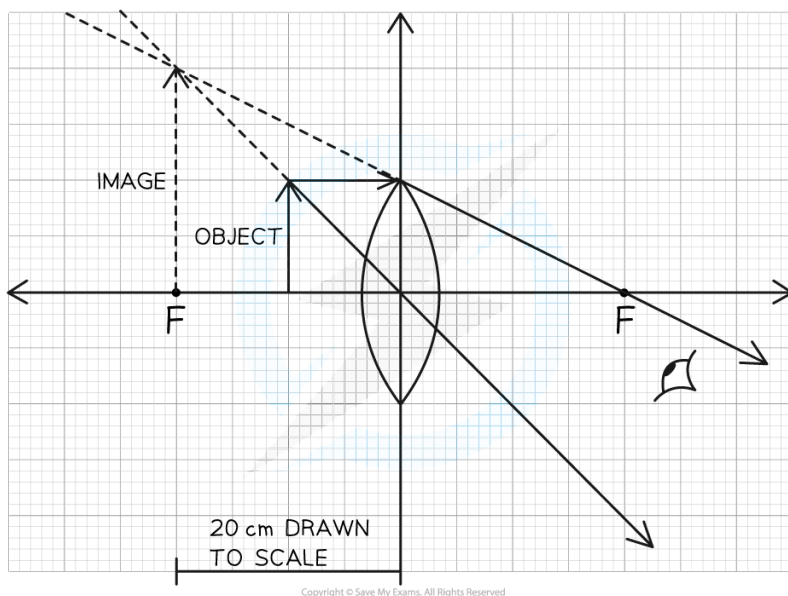


Your notes



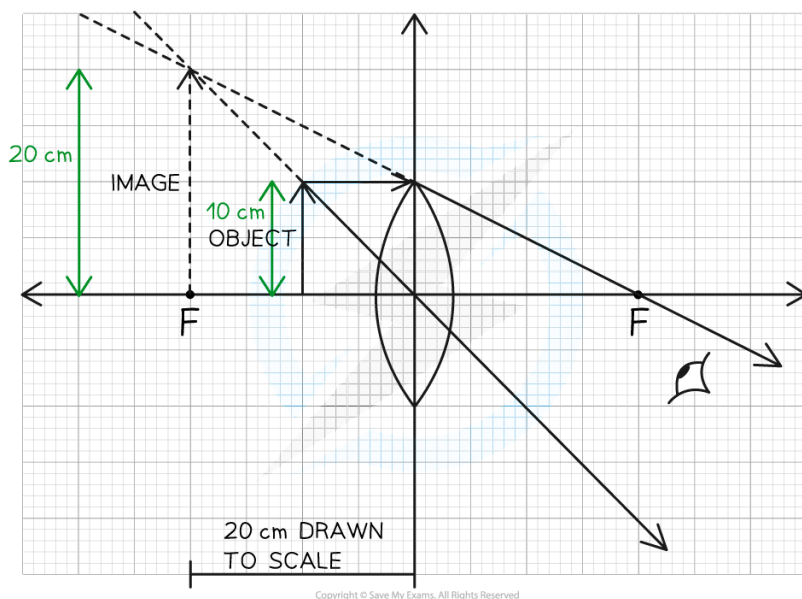
Worked Example

A person sees an image from a magnifying glass.



Calculate the magnification of this image. Clearly show your working on the diagram.

Answer:



Step 1: Measure the height of the object from the scale

The object is 10 cm

Step 2: Measure the height of image from the scale

The image is 20 cm

Step 3: Substitute values into the magnification equation

$$\text{Magnification} = \frac{\text{Image height}}{\text{Object height}} = \frac{20 \text{ cm}}{10 \text{ cm}} = 2$$



Examiner Tips and Tricks

The most common mistake with magnification calculations is to get the formula upside down.

Do a 'sanity check' by looking at the answer to make sure that magnified objects have got bigger ($m > 1$) and diminished ones smaller ($m < 1$).

Since we are working with ratios (so the units get cancelled out) this is one of those rare times when you don't need to convert everything to SI units, but do check that your units are all the same – for example all distances in cm.



Your notes



Plane Polarisation

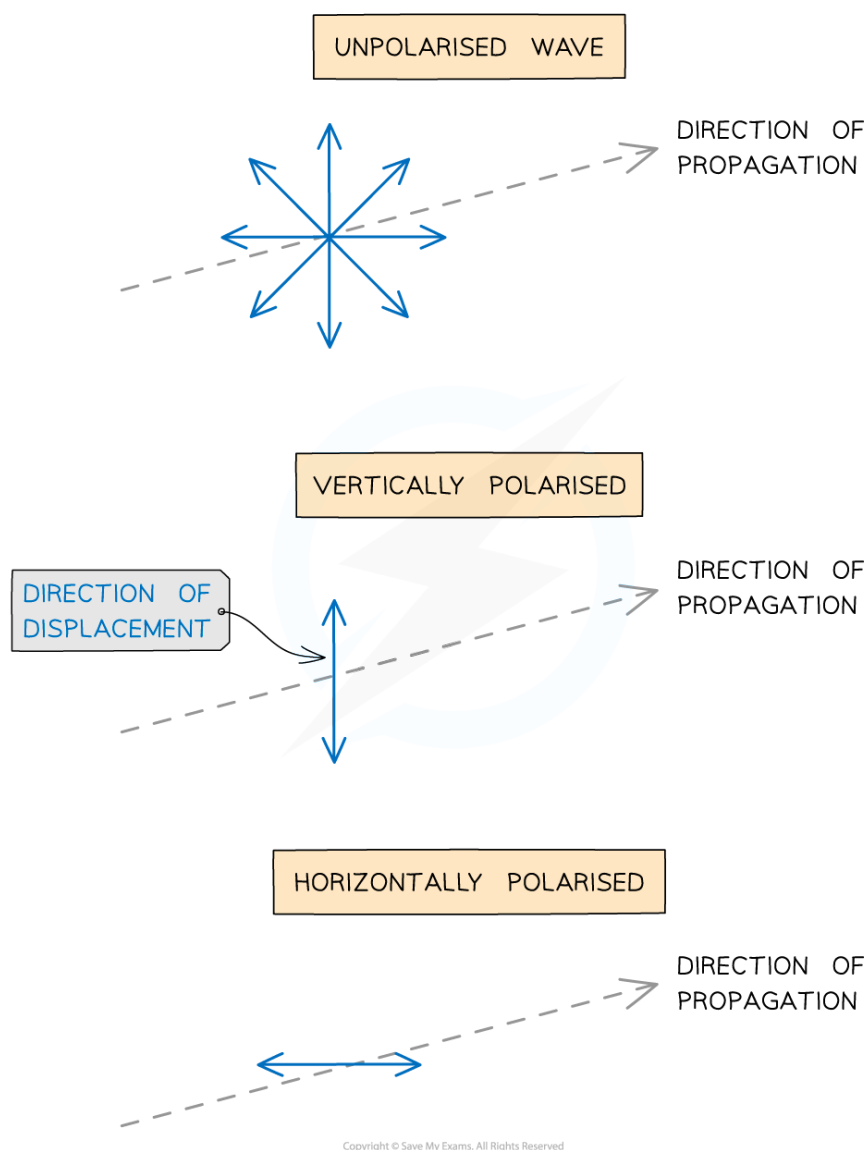
- Transverse waves can oscillate in any plane perpendicular to the direction of motion (and energy transfer) of the wave
- Such waves are said to be **unpolarised**
- **Polarisation** occurs when

Particles are only allowed to oscillate in one of the directions perpendicular to the direction of wave propagation

- When a transverse wave is polarised, its **electric field** is only allowed to oscillate in **one fixed plane** perpendicular to the direction of motion of the wave
 - For EM waves it is the **plane** of the **electric fields** oscillation that defines its **plane of polarisation**
- A transverse wave can be vertically polarised, horizontally polarised, or polarised in any direction in between



Your notes



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Diagram showing the displacement of unpolarised and polarised transverse waves

- Since longitudinal waves oscillate in the same direction as the direction of motion of the wave, **polarisation of longitudinal waves cannot occur**
- Methods of polarisation include polarising filters and reflection from a non-metallic plane surface

Polarising Filters

- Light waves can be polarised by making them pass through a **polarising filter** (also known as a **polariser**)
- The filter imposes its plane of polarisation on the incident light wave
- A polariser with a vertical **transmission axis** only allows vertical oscillations to be transmitted through the filter (**A**)

- If vertically polarised light is incident on a filter with a horizontal transmission axis, no transmission occurs (**B**), and the wave is blocked completely



Your notes

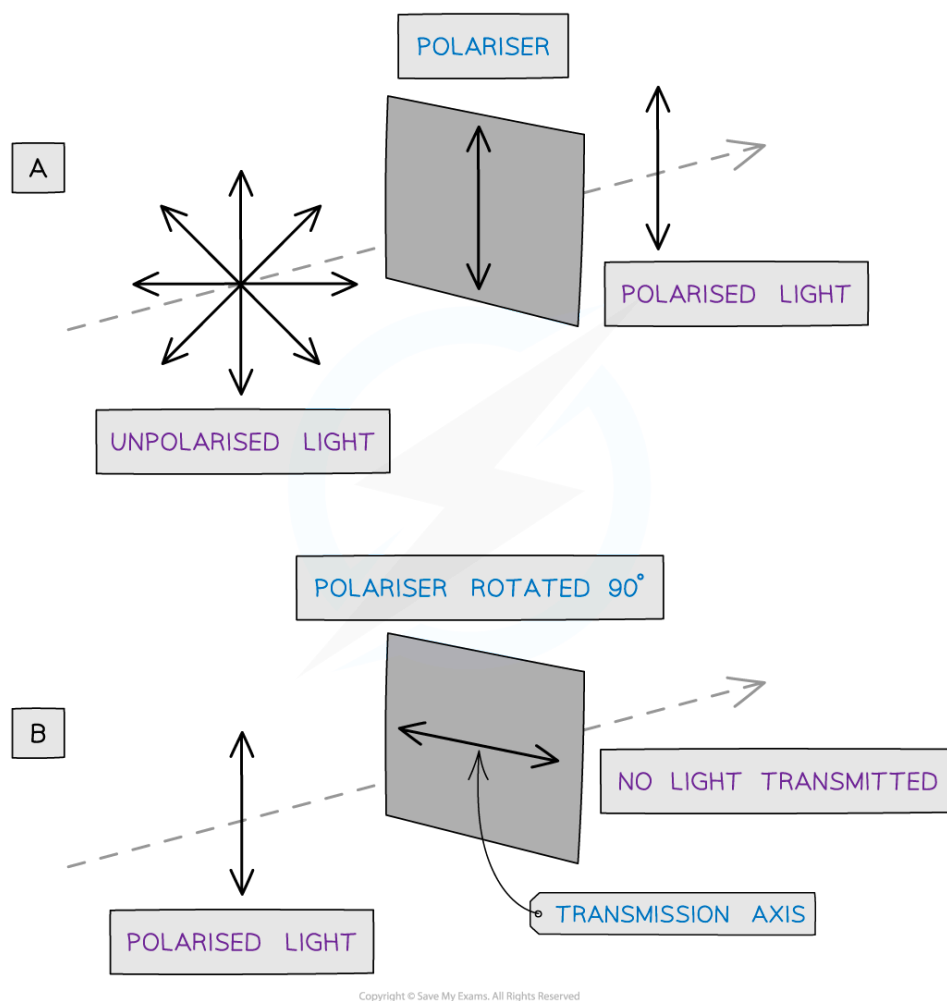


Diagram showing an unpolarised and polarised wave travelling through polarisers